

A sausage wrapped in newspaper

Why the cut edge of the paper unrolls into a sine wave

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About this version. This is a revised version of a school project. The idea, the experiment and the photographs are the original ones. The revision corrects the English and the notation, writes the derivation out in full, and replaces the visual comparison at the end with a measurement taken from the original scan (Section 5.1).

1 Introduction

I like noticing mathematics in ordinary things. One day I needed to take a piece of sausage to my grandmother. I wrapped the sausage in newspaper and cut through the paper and the sausage together (Figure 1). When I unfolded the newspaper at her home, its cut edge was a wave (Figure 2), and the wave looked very much like a sine curve.



Figure 1. The sausage, wrapped and cut.



Figure 2. The newspaper, unfolded.

2 Aim

To prove that the cut edge is a sinusoid, and to express its parameters through quantities that can be measured on the sausage with a ruler. (A cosine is a sine shifted by a quarter of a period, so “sinusoid” covers both.)

3 The model

Treat the dry-cured sausage as a cylinder of radius r whose axis is the z -axis, passing through O at the centre of the base and V at the centre of the top (Figure 3). The knife cut is a plane; I built it in GeoGebra through three points L , P and K . The axes are chosen so that the diameter AC of the base lies along the y -axis and the plane contains lines parallel to it. Then the plane rises only in the x direction, and its height above the base depends on x alone:

$$z = c + m x,$$

where m is the slope of the cut and c is its height above the centre O .

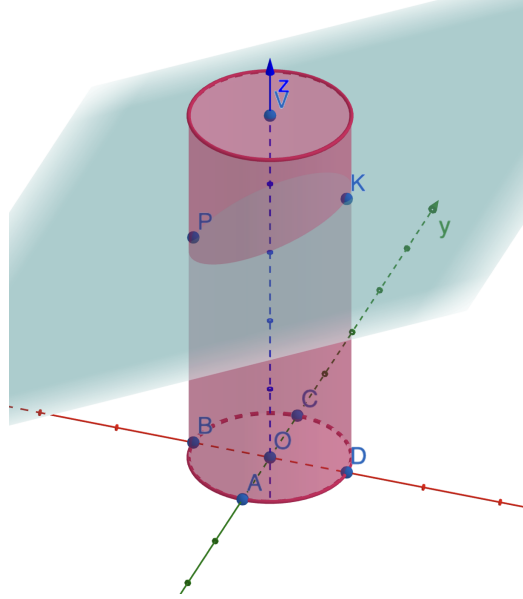


Figure 3. The sausage as a cylinder, cut by a plane through L , P and K .

4 The shape of the edge

Take a point G on the cut edge, and let x be the angle $\angle DOG$ at which it sits, measured from D in the base circle (Figure 4). The point of the base circle under G has coordinates $(r \cos x, r \sin x)$, so by the equation of the plane, G is at height

$$c + m r \cos x.$$

Heights depend only on the x -coordinate, so everything happens in the right triangle OFG : the height of G is set by the adjacent leg $OF = r \cos x$, not by the hypotenuse $OG = r$.

Now unroll the side of the cylinder, starting from D . It becomes a rectangle whose width is the circumference $2\pi r$, and the point at angle x lands at a distance $s = rx$ from D along the bottom edge. Substituting $x = s/r$, the height of the cut edge over the point s is

$$h(s) = c + m r \cos \frac{s}{r}.$$

The parameters. The highest point of the cut, K , is above D , where $x = 0$; the lowest, P , is above the opposite point B , where $x = \pi$ (Figure 5). Call their heights k and p . Then

$$k = c + mr, \quad p = c - mr \quad \implies \quad c = \frac{k + p}{2}, \quad mr = \frac{k - p}{2},$$

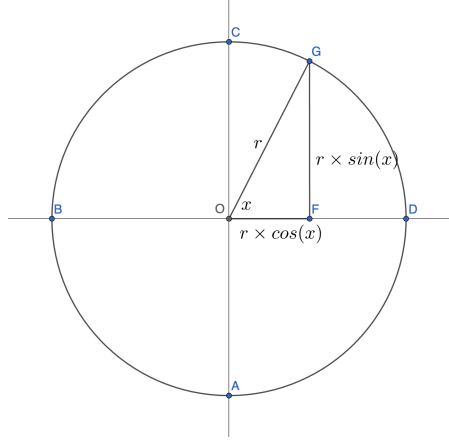


Figure 4. The base circle. The point at angle x from D has x -coordinate $OF = r \cos x$.

and therefore

$$h(s) = \frac{k-p}{2} \cos \frac{s}{r} + \frac{k+p}{2} \quad (1)$$

The edge is a sinusoid (Figure 6) with midline $\frac{k+p}{2}$, amplitude $\frac{k-p}{2}$ and period $2\pi r$. The period depends only on the thickness of the sausage; the height of the wave depends on how steeply it was cut, and a cut straight across ($k = p$) gives a straight edge. Two quick checks agree: a quarter of a turn from D , where $\cos \frac{s}{r} = 0$, the height is the average $\frac{k+p}{2}$, and at D itself, where $\cos \frac{s}{r} = 1$, it is k .

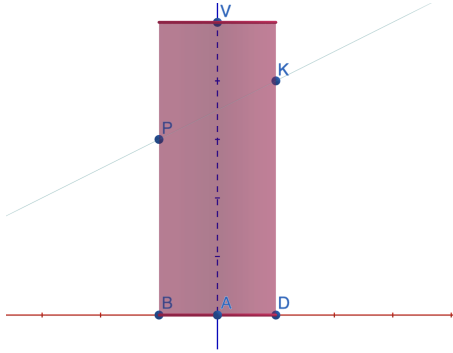


Figure 5. Side view: P and K are the lowest and highest points of the cut.

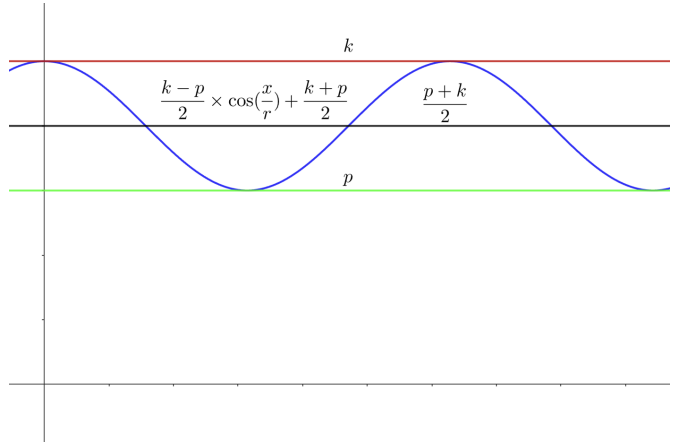


Figure 6. Formula (1): the wave between p and k .

5 The experiment

To check the formula I cut a real sausage. Its radius was $r = 1.7$ cm; after the cut, the lowest point of the cut was at $p = 5.2$ cm and the highest at $k = 7.8$ cm (Figure 7). Formula (1) then predicts

$$h(s) = \frac{7.8 - 5.2}{2} \cos \frac{s}{1.7} + \frac{7.8 + 5.2}{2} = 1.3 \cos \frac{s}{1.7} + 6.5,$$

a wave of period $2\pi \cdot 1.7 \approx 10.7$ cm. I cut and unwrapped the newspaper and scanned it on the school printer, to avoid the distortion a camera adds (Figure 8), then laid the predicted curve over the scan in GeoGebra (Figure 9).



Figure 7. The sausage and the newspaper before the cut.

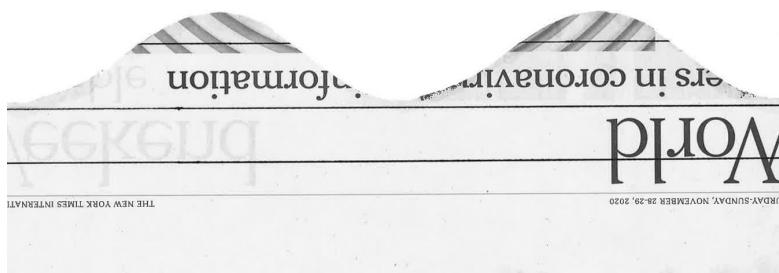


Figure 8. The unwrapped newspaper, scanned.

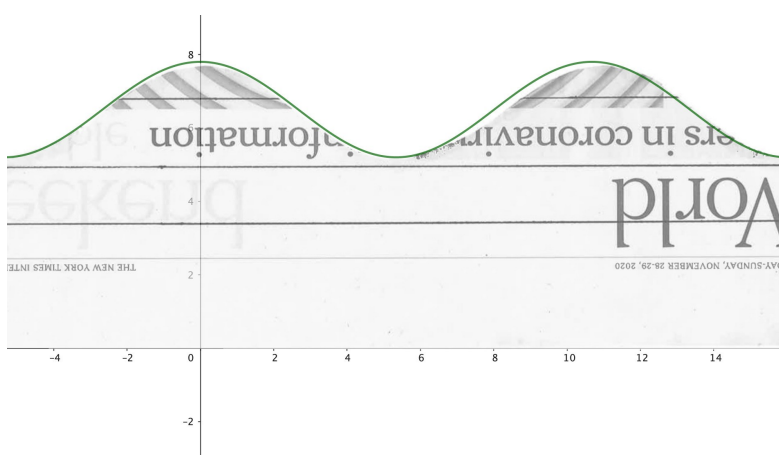


Figure 9. The predicted curve (green) over the scan, in GeoGebra.

5.1 Measuring the fit

The original comparison ended at “quite similar”. The scan allows a sharper test. Its absolute scale was not recorded, so the period in centimetres cannot be read from it; what can be read are the *shape* of the edge and the *ratio* of its height to its period, and neither depends on scale.

The edge was traced column by column. The paper is a shade darker than the scanner lid (a brightness of about 242 against 255), so in each column the edge is the first row where the paper begins. Of the 1,349 columns, 1,309 were used: the other 40 are where the crop of the scan cuts off the top of the second crest, or where a speck of dust sits above the edge. Each curve was fitted with its amplitude, period, position and midline free, and with the slight tilt of the scan free as well (it comes out at about 0.2°).

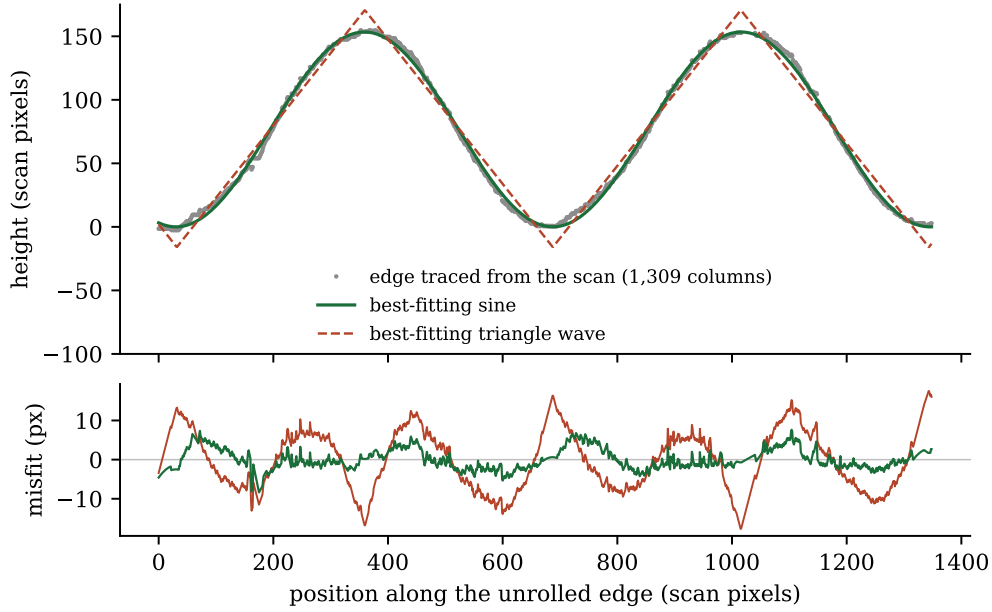


Figure 10. The traced edge, the best-fitting sine and the best-fitting triangle wave, with each curve’s misfit underneath.

	measured on the scan	predicted by (1)
height of the wave $\div$ period	0.233	$\frac{k-p}{2\pi r} = \frac{2.6}{10.68} = 0.243$
misfit of a sine (root mean square)	1.8% of the height	
misfit of a triangle wave, same freedom	4.8% of the height	

Table 1. The fit, in numbers.

Two things follow (Figure 10, Table 1).

The edge is a sine, and not just a wave. A sine follows it to within 1.8% of the wave’s height. A triangle wave, given exactly the same freedom, misses by 4.8%: its misfit swings in step with every crest and trough, and the sine’s is about a third as large.

The size of the wave matches the formula to 4%. The ratio of height to period is 0.233 on the scan against a predicted 0.243. If each ruler reading was good to about a millimetre, then $k - p = 2.6 \pm 0.14$ cm and $r = 1.7 \pm 0.05$ cm (half of a diameter read to a millimetre), which leaves the prediction itself uncertain by about 6%; a gap of 4% is inside that. Changing

how the edge is traced (24 combinations of brightness threshold and smoothing) moves the measured ratio only between 0.233 and 0.234.

The ratio has one more advantage: it does not care that the newspaper was wound in several layers. An outer layer has a larger radius, which stretches both the period and the height of its wave by the same factor, so the ratio is the same for every layer. A check of the period alone would need the radius of the paper, not of the sausage. What remains of the gap has natural causes: a knife cut is not a perfect plane, and the paper does not sit perfectly tight on the sausage.

6 Conclusion

The cut edge of a newspaper wrapped around a cylindrical sausage unrolls into the sinusoid

$$h(s) = \frac{k-p}{2} \cos \frac{s}{r} + \frac{k+p}{2},$$

where r is the radius of the sausage, s is the distance along the unrolled edge, and p and k are the lowest and highest points of the cut. On a real sausage the edge follows a sine to within 1.8% of its height, and the size of the wave matches the formula to within the precision of a ruler.

7 Further questions

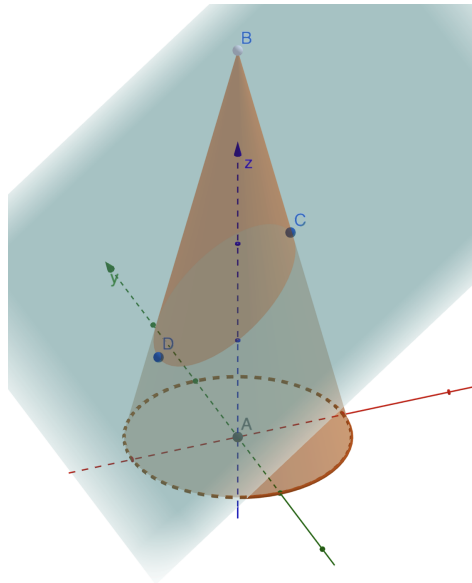


Figure 11. A cone cut by a plane.

The first extension is to replace the cylinder with a cone (Figure 11). A cone unrolls into a sector of a circle rather than a rectangle, so the question becomes what the cut edge looks like on that sector.

The second is a sausage with an oval cross-section. I tried this case and did not solve it. The difficulty is that the distance along the perimeter of an ellipse has no elementary formula (it is an elliptic integral), so the position s on the unrolled edge is no longer a simple multiple of the angle.